ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 4.01: Logarithmic & Exponential Functions

Q1. Solve for x : log_{3}x + log_{3}4 = log_{3}16 .

[x] A: 4
[ ] B: 6
[ ] C: 8
[ ] D: 2

Solution: Using log laws: log_{3}4x = log_{3}16 implies 4x = 16 implies x = 4 .

Q2. Solve for x : log_{4}x + log_{4}4 = log_{4}36 .

[ ] A: 3
[ ] B: 18
[ ] C: 11
[x] D: 9

Solution: Using log laws: log_{4}4x = log_{4}36 implies 4x = 36 implies x = 9 .

Q3. Solve for x : log_{5}x + log_{5}4 = log_{5}64 .

[ ] A: 4
[ ] B: 32
[ ] C: 18
[x] D: 16

Solution: Using log laws: log_{5}4x = log_{5}64 implies 4x = 64 implies x = 16 .

Q4. Solve for x : log_{6}x + log_{6}4 = log_{6}100 .

[x] A: 25
[ ] B: 27
[ ] C: 5
[ ] D: 50

Solution: Using log laws: log_{6}4x = log_{6}100 implies 4x = 100 implies x = 25 .

Q5. Solve for x : log_{7}x + log_{7}4 = log_{7}144 .

[ ] A: 38
[x] B: 36
[ ] C: 72
[ ] D: 6

Solution: Using log laws: log_{7}4x = log_{7}144 implies 4x = 144 implies x = 36 .

Q6. Solve for x : log_{8}x + log_{8}4 = log_{8}196 .

[ ] A: 7
[ ] B: 98
[x] C: 49
[ ] D: 51

Solution: Using log laws: log_{8}4x = log_{8}196 implies 4x = 196 implies x = 49 .

Q7. Solve for x : log_{9}x + log_{9}4 = log_{9}256 .

[ ] A: 66
[ ] B: 128
[x] C: 64
[ ] D: 8

Solution: Using log laws: log_{9}4x = log_{9}256 implies 4x = 256 implies x = 64 .

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Q8. Solve for x : log_{10}x + log_{10}4 = log_{10}324 .

[ ] A: 9

[ ] B: 162

[ ] C: 83

[x] D: 81

Solution: Using log laws: log_{10}4x = log_{10}324 implies 4x = 324 implies x = 81 .